

Physics and Equality Constrained Artificial Neural Networks For Options Pricing

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1 Abstract

This paper presents a Physics and Equality Constrained Artificial Neural Network (PECANN) for pricing European-style call options under the Black-Scholes model. Standard Physics-Informed Neural Networks (PINNs) applied to option pricing suffer from two related weaknesses: poor convergence caused by disparate loss-term magnitudes, and a dimensionality problem in which a trained network is valid for only a single combination of strike price, volatility, and risk-free rate. This work addresses the second issue by non-dimensionalising the Black-Scholes equation with respect to strike price and time to maturity, training a single network across ranges of $\hat{S}$, $\hat{t}$, r , and σ to obtain, in effect, a universal European call pricing model. The first issue is addressed by reformulating the PINN loss into a PECANN using the Augmented Lagrangian Method, treating the boundary and terminal conditions as hard equality constraints while minimising the PDE residual directly. An adaptive collocation refinement scheme, using KDTree based nearest neighbour re-sampling, is further implemented to concentrate training points in regions of high PDE residual. Relative to a baseline PINN, the PECANN reduces boundary and data loss by approximately three orders of magnitude and improves RMSE against the closed-form solution by up to two orders of magnitude, with adaptive refinement only able to offer a marginal advancement on this result which is found to be statistically indistinguishable from noise. The resulting model is approximately 280 times more accurate at maturity than a comparable gated PINN architecture reported in a PINN option pricing paper, though this is not an absolute comparison between papers and is currently restricted to European call options with a fixed one-year maturity. The model is least accurate near the terminal payoff due to architecture constraints.

2 Introduction

Closed-form solutions to the Black-Scholes equation exist for European options, but the majority of instruments traded in practice such as American options, multi-asset rainbow options, or products under more realistic volatility dynamics such as Heston, have no closed-form solution and must be valued with numerical methods. Traditional approaches, including finite-difference and Monte Carlo methods, are well established but scale poorly as the dimensionality of the problem increases, since each additional state

variable typically adds a dimension to a grid or a corresponding increase in required sample paths.

Physics-Informed Neural Networks (PINNs), introduced by Raissi et al [1], offer an alternative approach to solving partial differential equations. Instead, a neural network is trained to approximate the solution directly, with derivatives of the network's output with respect to its inputs computed via automatic differentiation and substituted into the governing PDE to form a residual loss. This residual loss, combined with losses against boundary, terminal, and any available data conditions, allows the network to be trained without ever discretising the domain, making PINNs a natural candidate for pricing problems where grid-based methods become computationally expensive at higher dimensions.

Applying this framework directly to Black-Scholes reveals two practical limitations. Firstly, a PINN trained for one set of parameters: strike price, volatility, and risk-free rate, is only valid for that combination; pricing a new option requires retraining, at a computational cost that may exceed that of a standard solver. Secondly, the composite PINN loss function combines terms of very different magnitudes (data, boundary condition, and PDE residual losses), which distorts the loss landscape and degrades convergence, particularly for the boundary and terminal conditions that significantly impact the quality of the solution.

This paper addresses both limitations. The Black-Scholes equation is non-dimensionalised with respect to strike price and time to maturity, and the network is extended to accept risk-free rate and volatility as additional inputs alongside asset price and time, producing a single model valid across a continuous range of option parameters rather than one strike and volatility. To address the convergence issue, the PINN loss is reformulated as a Physics and Equality Constrained Artificial Neural Network (PECANN), following Basir and Senocak [4, 5], in which the boundary and terminal condition losses are treated as hard equality constraints via the Augmented Lagrangian Method while the PDE residual is minimised directly. An adaptive collocation point refinement scheme is also introduced, using a KDTree based nearest neighbour construction to concentrate additional training points around regions of high PDE residual during training.

3 Theory & Governing Equations

The one-dimensional Black-Scholes equation is defined as below [6].

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0 \quad (1)$$

The terms of the equation are defined as follows. V , the option value, t , the time at which the option valuation takes place, σ , the annualised volatility, S , the underlying asset price and r , the risk free rate.

The Black-Scholes equation can be represented with PINNs and PECANNs alike, but faces the issue of dimensionality. After training the model is only valid for a specific strike price, K , annualised volatility, σ , risk free rate, r , and range of asset prices, S . This presents the obvious issue in pricing a variety of options with different asset and strike prices as well as risk free rates and volatilities. A new model would need to be trained for a change in any of the previously stated parameters at a likely larger computational cost than that of a standard solver approach.

3.1 Non-Dimensional Black-Scholes Equation Derivation

To avoid the issue described previously, we can nondimensionalise the governing variables, introducing the below scaled quantities.

$$\hat{V} = \frac{V}{K}, \quad \hat{t} = \frac{t}{T}, \quad \hat{S} = \frac{S}{K}$$

Here, $\hat{V}$ denotes the option value normalised by the strike price K , $\hat{t}$ is the time variable normalised by the time to maturity T , and $\hat{S}$ is the underlying asset price normalised by the strike. All parameters with units of currency are normalised by the strike price and parameters with units of time are normalised by the time to maturity.

Firstly, substituting the non-dimensional option value, $\hat{V}$, into the Black Scholes equation we obtain the following.

$$\frac{\partial(K\hat{V})}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2(K\hat{V})}{\partial S^2} + rS \frac{\partial(K\hat{V})}{\partial S} - r(K\hat{V}) = 0$$

This can be simplified by dividing through by K to the below.

$$\frac{\partial \hat{V}}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 \hat{V}}{\partial S^2} + rS \frac{\partial \hat{V}}{\partial S} - r\hat{V} = 0 \quad (2)$$

Using the chain rule we can define the following substitutions for each derivative.

$$\frac{\partial \hat{V}}{\partial t} = \frac{\partial \hat{V}}{\partial \hat{t}} \frac{\partial \hat{t}}{\partial t} = \frac{\partial \hat{V}}{\partial \hat{t}} \frac{1}{T}$$

$$\frac{\partial \hat{V}}{\partial S} = \frac{\partial \hat{V}}{\partial \hat{S}} \frac{\partial \hat{S}}{\partial S} = \frac{\partial \hat{V}}{\partial \hat{S}} \frac{1}{K}$$

$$\frac{\partial^2 \hat{V}}{\partial S^2} = \frac{\partial^2 \hat{V}}{\partial \hat{S}^2} \frac{\partial \hat{S}^2}{\partial S^2} = \frac{\partial^2 \hat{V}}{\partial \hat{S}^2} \frac{1}{K^2}$$

Substituting the above expressions into equation 2 and recalling the definition $S = \hat{S}K$, we obtain the below.

$$\frac{\partial \hat{V}}{\partial \hat{t}} \frac{1}{T} + \frac{1}{2} \sigma^2 \hat{S}^2 K^2 \frac{\partial^2 \hat{V}}{\partial \hat{S}^2} \frac{1}{K^2} + r\hat{S}K \frac{\partial \hat{V}}{\partial \hat{S}} \frac{1}{K} - r\hat{V} = 0$$

This equation can then be simplified to the final partially non-dimensional Black-Scholes equation below.

$$\frac{\partial \hat{V}}{\partial \hat{t}} \frac{1}{T} + \frac{1}{2} \sigma^2 \hat{S}^2 \frac{\partial^2 \hat{V}}{\partial \hat{S}^2} + r\hat{S} \frac{\partial \hat{V}}{\partial \hat{S}} - r\hat{V} = 0 \quad (3)$$

The obvious caveat with this derivation is that T , the time to maturity of the option, is still dimensional. There is less variability in the maturities between options than asset value and strike price. For example, two options can often have the same time to maturity but rarely have the same asset value and strike, making the non-dimensionalisation of the asset and option price more essential. I go on to discuss solutions to this issue in the future work section, but in this paper the time to maturity is fixed at 1 year for all results presented.

3.2 Boundary Conditions

The model was trained to price European style call options. The boundary conditions for a European call option are defined below.

$$\hat{V}(0, \hat{t}) = 0 \quad \hat{V}(\hat{S}, \hat{t}) = \hat{S} - e^{-rT(1-\hat{t})} \text{ as } \hat{S} \rightarrow \infty$$

The $\hat{S}$ grid on which the call option value is evaluated ranges from zero to $\hat{S}_{max}$, which for this model is defined as 3 times the strike price or an $\hat{S}$ value of 3. The strike price, K , is normalised by itself taking a value of 1, and hence not seen in the $\hat{S} = \hat{S}_{max}$ boundary condition above or later in the terminal condition.

3.3 Terminal Condition

For a European style call option, the terminal condition is defined below.

$$\hat{V} = \max(\hat{S} - 1, 0)$$

3.4 Physics Informed Neural Network (PINN)

The PINN was first theorised and implemented by Raissi et al [1] as an improved method of using neural networks to predict the solution of non-linear partial differential equations. Whereas a traditional neural network is solely penalised for the error of the models prediction to a set of output data points as a function of the corresponding input data points, a PINN additionally penalises the model for error to boundary, initial, and terminal conditions as well as unphysical solutions at random collocation points within the computational domain. The derivatives of the network's inputs, the inputs to the partial differential equation, can be computed with respect to the network's output, the solution of the differential equation, using automatic differentiation. These derivatives can then be directly substituted into the governing equation alongside all other known terms to compute the error to the exact physical solution at collocation points. In the case of Black-Scholes, the derivatives $\frac{\partial V}{\partial t}$, $\frac{\partial V}{\partial S}$, and $\frac{\partial^2 V}{\partial S^2}$ are computed at a low computational cost via automatic differentiation. The founding paper noted an improvement in the ability of neural networks to represent partial differential equations using this three-term loss-equation structure. In this paper known issues with the PINN

method are identified and a Physics and Equality Constrained Artificial Neural Network (PECANN) is implemented.

4 Methodology

4.1 Network Architecture

A multi-layer perceptron structure (MLP) was utilised, with 4 inputs to the model; non-dimensional asset price, $\hat{S}$, non-dimensional time, $\hat{t}$, the risk free rate, r , and the annualised volatility, σ . The network featured a single linear output layer for the non-dimensional option price, $\hat{V}$. The deeper network consisted of 3 hidden layers of 32 perceptrons each. A hyperbolic tangent activation function was selected as a PINN requires a smooth and continuous function in order to be differentiable via automatic differentiation. A rectified linear unit activation function has an undefined second order derivative at its kink, and so is not able to provide the $\frac{\partial^2 \hat{V}}{\partial \hat{S}^2}$ term at collocation points, required by the Black-Scholes equation.

4.2 PINN Loss Formulation

Firstly, the loss function of a standard PINN is defined below.

$$L_{Total}(\theta) = L_{Data}(\theta) + L_{PDE}(\theta) + L_{BC}(\theta)$$

Here L , represents the mean squared error (MSE) for each source, the loss to the data, governing equation at collocation points, and boundary conditions. The parameter, θ , represents all of the weights and biases that can be optimised to minimise the loss terms. Basir and Senocak [4, 5], state that an issue with this loss formulation for a PINN is the disparity in magnitudes of error between the 3 different terms. This affects the smoothness of the loss surface and subsequently the ability of the optimiser to find a minimum. To mitigate this effect each term was then normalised by it's first value at epoch 1. The new formulation is seen below.

$$L_{Total}(\theta) = \frac{L_{Data}(\theta)}{L_{Data_0}(\theta)} + \frac{L_{PDE}(\theta)}{L_{PDE_0}(\theta)} + \frac{L_{BC}(\theta)}{L_{BC_0}(\theta)}$$

It is worth noting some results from the above PINN loss formulation, seen below in figures 1 and 2, before progressing to the PECANN loss formulation.

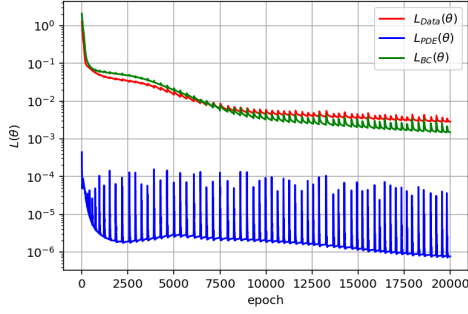


Figure 1: PINN Loss

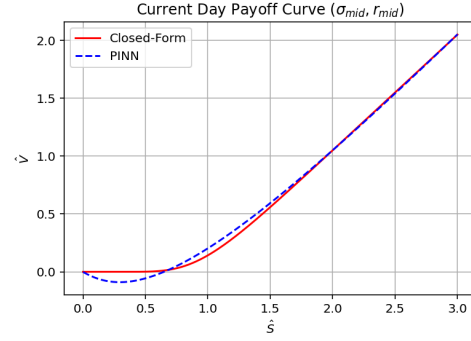


Figure 2: PINN Payoff Curve

In figure 1 we can see an obvious disparity in the magnitude of the error at the collocation points (PDE loss) to the error to the data and boundary conditions. The unstable behaviour of each loss term throughout the epochs is indicative of the optimising algorithm, Adam in this case, struggling to locate a minimum. The payoff curve, figure 2, shows poor agreement between the PINN and exact closed-form result, particularly in the region between $\hat{s}$ values of 0 and 1. A large boundary condition loss can propagate into the domain and affect the quality of the overall solution considerably.

The above result led to the implementation of a PECANN over the original PINN. A PECANN uses the Augmented Lagrangian Method (ALM) to treat one or many terms as hard equality constraints, whilst minimising another. In this case we can use an ALM formulation to constrain the boundary condition and data losses to be zero whilst minimising the loss at the collocation points, with the aim of improving the convergence of the data and boundary condition losses and so the payoff curve solution.

4.3 PECANN Loss Formulation

For a PECANN the loss formulation using ALM becomes the following.

$$L_{Total}(\theta, \lambda_{BC}, \lambda_{Data}, \mu_{BC}, \mu_{Data}) = L_{PDE}(\theta) + \lambda_{BC}^{\top} R_{BC}(\theta) + \frac{\mu_{BC}}{2} \|R_{BC}(\theta)\|^2 + \lambda_{Data}^{\top} R_{Data}(\theta) + \frac{\mu_{Data}}{2} \|R_{Data}(\theta)\|^2 \quad (4)$$

Here $L_{PDE}(\theta)$ is the MSE to the governing equation at the collocation points as before, $R_{BC}(\theta)$ and $R_{Data}(\theta)$ are the raw errors to the boundary conditions and

data and are column vectors of length N , where N is the number of data or boundary condition points. The λ terms are column vectors of length N , where N is defined as before, and both μ terms take a scalar value. The parameters λ_{BC} , λ_{Data} , μ_{BC} , and μ_{Data} are defined in the series of equations below.

$$\lambda_{BC}^{k+1} = \lambda_{BC}^k + \mu_{BC}^k R_{BC}(\theta_k)$$

$$\lambda_{Data}^{k+1} = \lambda_{Data}^k + \mu_{Data}^k R_{Data}(\theta_k)$$

where...

$$\mu_{Data}^{k+1} = \begin{cases} \beta \mu_{Data}^k & \text{if } ||R_{Data}(\theta_k)|| > \gamma ||R_{Data}(\theta_{k-1})|| \\ \mu_{Data}^k & \text{otherwise} \end{cases}$$

$$\mu_{BC}^{k+1} = \begin{cases} \beta \mu_{BC}^k & \text{if } ||R_{BC}(\theta_k)|| > \gamma ||R_{BC}(\theta_{k-1})|| \\ \mu_{BC}^k & \text{otherwise} \end{cases}$$

The above Lagrange multiplier, λ , and penalty parameter, μ , terms are updated every k epochs. The equations introduce a growth term, β , and a residual ratio term, γ , which must be chosen carefully to ensure smooth convergence during training. This formulation checks for the data and boundary condition error individually if there has been an adequate decrease every k epochs, determined by the term γ , and then adjusts the μ parameter by multiplying by the growth factor, β , if this criterion has not been met. This serves to more heavily punish the network for errors to the data, as well as boundary and terminal conditions. Therefore, inadequate error reductions for either constrained variable is met with a larger total loss, L_{Total} .

4.4 Optimiser Strategy

An Adam optimiser with a learning rate of 1e-3 and a weight decay of 1e-5 was used for the first 4000 epochs, for its low computational cost to begin convergence. This was followed by an Limited-memory Broyden–Fletcher–Goldfarb–Shanno (LBFGS) optimiser with a Strong Wolfe line search, a learning rate of 1.0, and a maximum number of 10 inner iterations with a 20 outer iteration history.

This provided stronger convergence but at a higher computational cost in the final 4000 epochs of a total 8000 epochs in training.

4.5 Adaptive Collocation Point Refinement

To improve convergence further and minimise error to the governing equation at collocation points a refinement strategy was devised. A schematic of the strategy is seen in figure 3.

The user has the ability to control the threshold at which refinement points are added in the neighbouring region, the number, n , of points added, as well as the maximum total collocation points before further refinement is prevented. The user can also control the period, i , in which the refinement takes place.

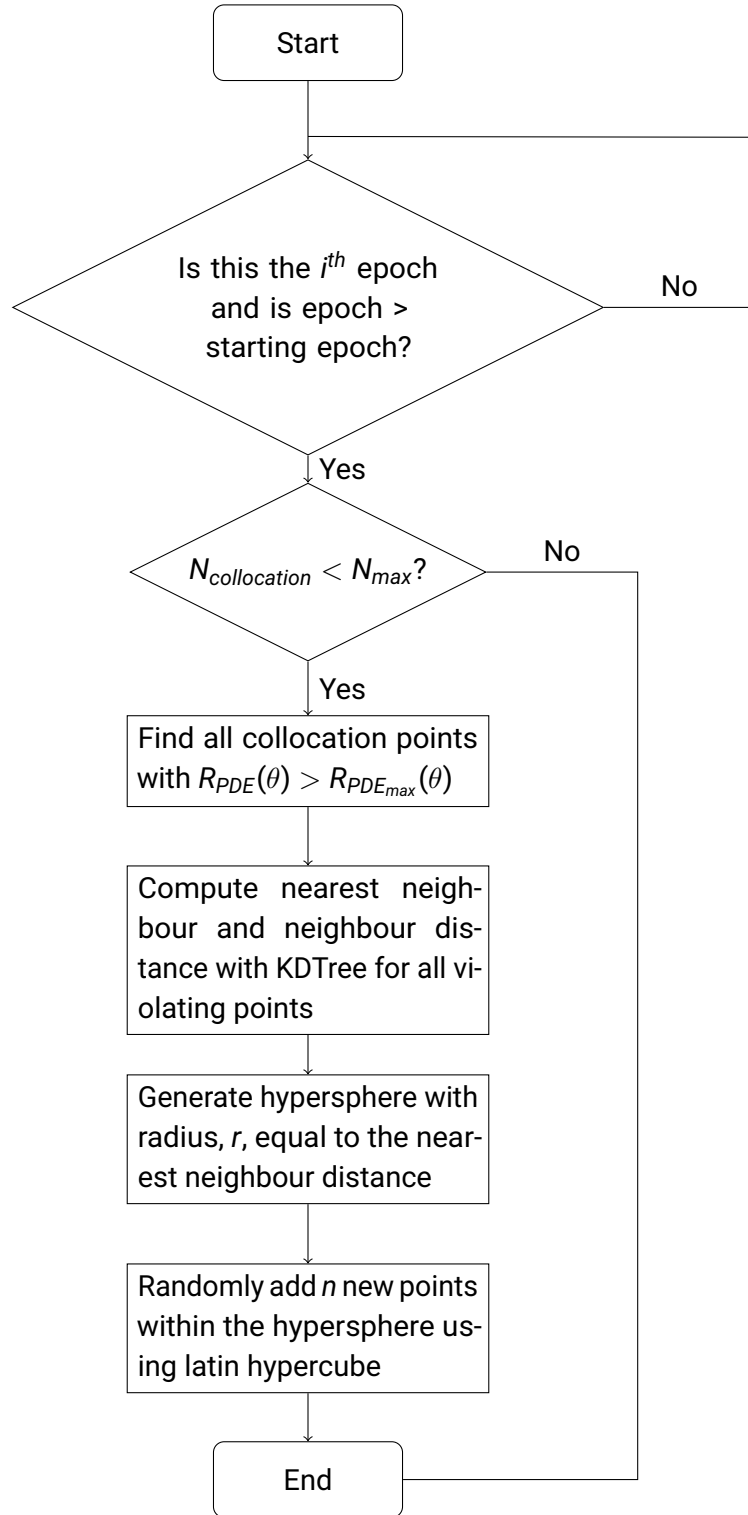


Figure 3: Adaptive Collocation Point Refinement Strategy Flowchart

5 Implementation

To train the network 2000 random data points were created using Latin hypercube, each with a unique combination of $\hat{S}$, $\hat{t}$, r , and σ values. The $\hat{S}$ values ranged from 0 to 3, the $\hat{t}$ values from 0 to 1, the r values from 0 to 0.1, and the σ values from 0 to 0.6. The same process was repeated again for a random set of 2000 collocation points throughout the domain. A uniformly spaced set of 100 points was generated for the boundary condition at $\hat{S} = 0$, at $\hat{S} = \hat{S}_{max}$, and for the terminal condition at $\hat{t} = 1$, for a total of 300 boundary condition points. Corresponding non-dimensional option prices, $\hat{V}$, were calculated using the closed-form call equation and boundary or terminal condition equations for the data and boundary condition points respectively. It should be noted that all experiments were run on a laptop with an Intel CPU (4 cores / 8 threads), 17 GB RAM, and integrated graphics. The root-mean-squared error (RMSE) was evaluated against the closed-form solution at the present day and, subsequently in figure 11, across non-dimensional time. All RMSE values were computed at 200 uniformly spaced points spanning the full $\hat{S}$ domain.

A table containing the list of hyper-parameters required by a PECANN is seen below.

Hyper-Parameter	Value
Update Period (epochs), k	50
Growth Rate, β	1.05
Residual Ratio Threshold, γ	0.5
Maximum Penalty Parameter, μ_{max}	1000
Adaptive Collocation Refinement	Off

Table 1: Table of PECANN Hyper-Parameters

6 Validation & Results

6.1 PECANN Result

The loss curve as well as the value of the multiplier parameter and residual ratio through training are seen in figure 4 below.

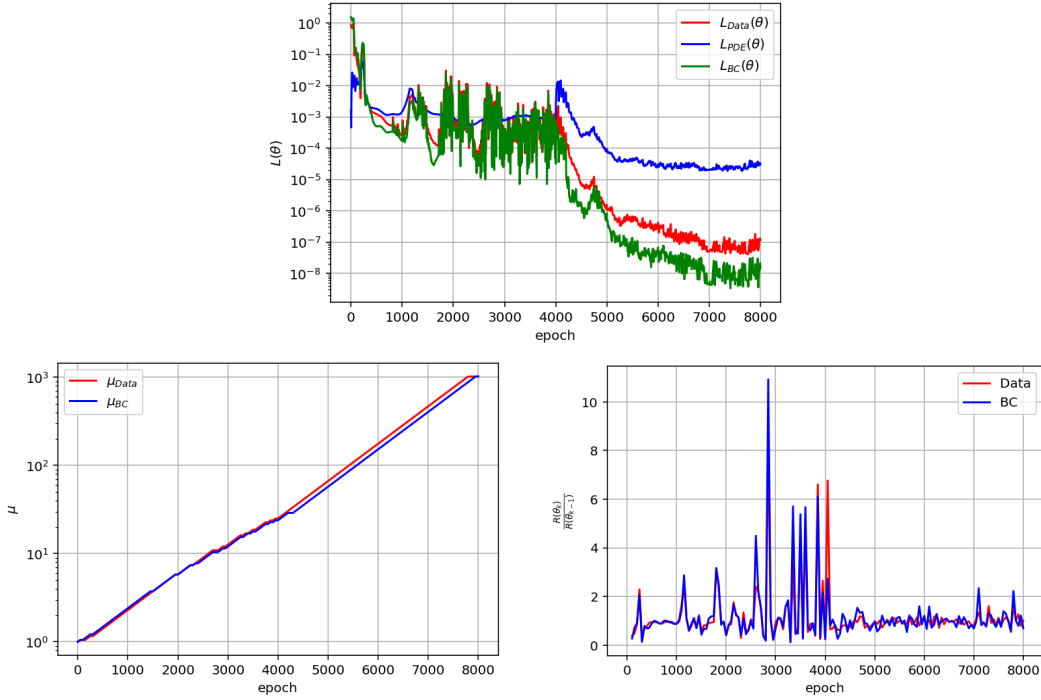


Figure 4: Top: PECANN loss against epochs. Bottom: evolution of the multiplier parameter μ and residual ratio γ across training.

We can see from figure 4 when compared to figure 1 that the loss to the boundary condition points and data is decreased by approximately 3 orders of magnitude. At the same time the loss as the collocation points (PDE) is around 2 orders of magnitude larger than the PINN result previously but still reaches a loss close to $1e-5$. The residual ratio plot demonstrates some instability in convergence with the Adam optimiser during the first 4000 epochs. The LBFGS optimiser in contrast provides a much more stable and consistent residual ratio through training, and consequentially the reduction of the losses in the final 4000 epochs is significant. There is a 3 order of magnitude difference between the initial losses of the collocation and data or boundary condition points. Such a difference in scale creates a distorted,

non-convex loss surface and negatively impacts the optimisation algorithm [4].

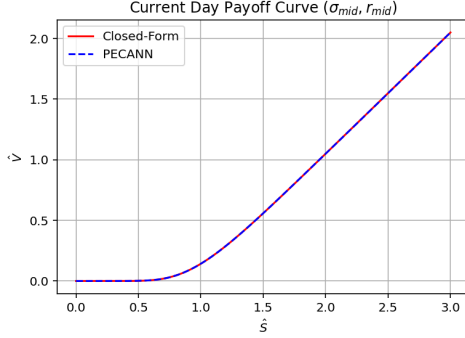


Figure 5: PECANN Payoff Curve

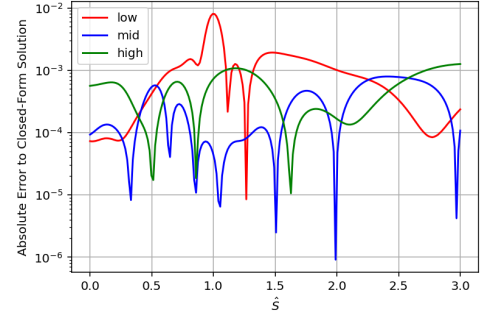


Figure 6: Absolute Error Against $\hat{S}$

Point Location	σ	r
Low	0.05	0.01
Mid	0.3	0.05
High	0.55	0.09

Table 2: Table of Low, Mid, and High σ and r Values

Figure 5 presents the payoff curve and figure 6 the absolute error to the closed-form solution for a European call option. The absolute error is calculated for 3 different combinations of σ and r values, at a low, mid, and high point of the learned range of values as seen in table 2. We can see in contrast to the PINN payoff curve presented in figure 2, the accuracy of the solution is significantly better, with an indecipherable difference between the closed-form and PECANN solution. The payoff curve is calculated for present day and an option with a time to maturity, T , of 1 year.

6.2 Adaptive PECANN Result

In an effort to further improve the quality of the PECANN results presented above, the adaptive refinement algorithm seen in figure 3 was implemented with the parameters seen in table 3. The results are then presented below.

Parameter	Value
Refinement Starting Epoch	5000
Residual Threshold	5e-5
Maximum Collocation Points	25000
Points Added Per Adaptation	3
Adaptation Period	500

Table 3: Table of Adaptive Refinement Parameters and Values

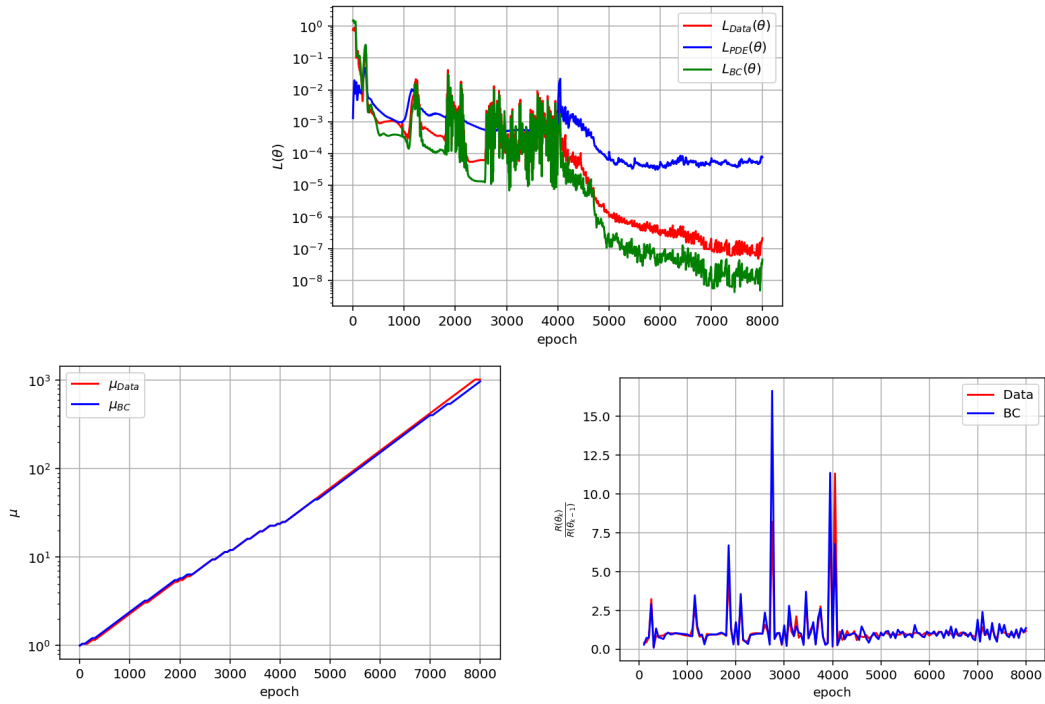


Figure 7: Top: Adaptive PECANN loss against epochs. Bottom: evolution of the multiplier parameter μ and residual ratio γ across training.

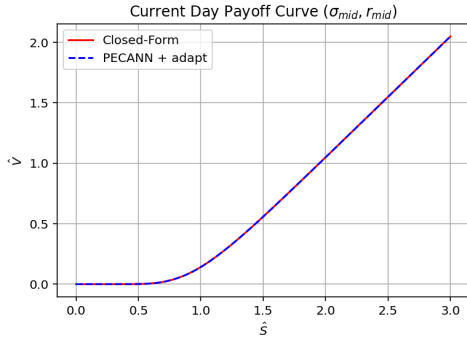


Figure 8: Adaptive PECANN Payoff

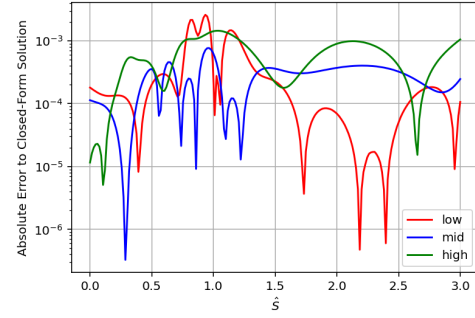


Figure 9: Absolute Error Against $\hat{S}$

The residual at the collocation points throughout the domain and the corresponding refinement process that took place are demonstrated below in figure 10.

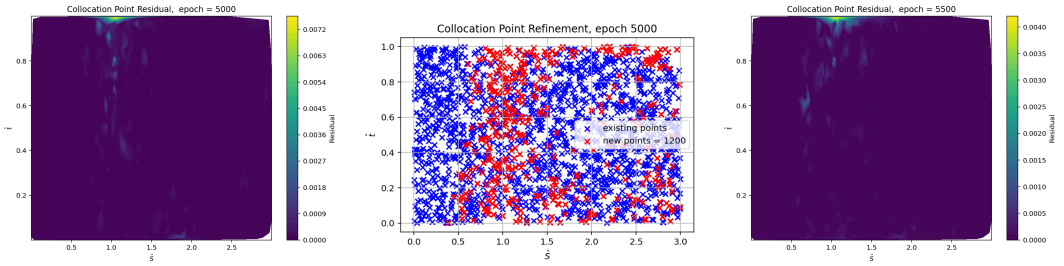


Figure 10: From left to right: the PDE residual at epoch 5000, the corresponding collocation points added, and the PDE residual at epoch 5500

Again, the payoff curve is indecipherable compared to the closed form solution for the adaptive PECANN. Inspecting figure 9, we can see that the absolute error has a reduced peak for the low parameters around an $\hat{S}$ value of 1 compared to the standard PECANN. The adaptive refinement reduces the maximum collocation-point residual from approximately 0.0072 at epoch 5000 to 0.0040 at epoch 5500, indicating a meaningful improvement in the local accuracy of the solution.

6.3 Comparison of Results

A table comparing the PINN, PECANN, and adaptive PECANN results is seen below.

Model	Low RMSE	Mid RMSE	High RMSE	Training Time (s)	Epochs	Collocation Added
PINN	5.369×10^{-2}	4.075×10^{-2}	5.046×10^{-2}	188.9	20,000	–
PECANN	1.708×10^{-3}	3.932×10^{-4}	6.308×10^{-4}	506.6	8,000	–
Adaptive PECANN	5.963×10^{-4}	3.011×10^{-4}	7.156×10^{-4}	1274.7	8,000	12,912

Table 4: Comparison of PINN, PECANN, and Adaptive PECANN Performance Metrics at Present Day, $\hat{t} = 0$.

The adaptive PECANN provides the lowest RMSE across the range of input parameters when tested, but at 2.5 times the computational cost to the standard PECANN. Whilst this cost may not ordinarily be justifiable, since the network is non-dimensional the training time becomes less of a driving factor in favour of accuracy. The same network can price a range of options of the same type, not just a singular option, preventing the need for retraining. The adaptive algorithm refinement, seen in figure 10, targets collocation points at $\hat{S}$ values close to 1 in particular, where the payoff curve has high curvature. This behaviour is useful, as smooth activation functions typically exhibit difficulty representing regions of high curvature. By concentrating refinement in these areas, the algorithm enhances local resolution of the solution, contributing directly to the reduction in absolute error observed for the low-parameter case near $\hat{S} = 1$. However, subsequently in section 6.4 this is shown to be an outlier result, not holding up under seed variance testing.

The quality of these results can be compared to previous research into PINNs for option pricing completed by Dhiman & Hu [7]. They were able to achieve an RMSE of 0.228 to the true closed-form value for a European call option at payoff ($\hat{t}=1$) using a PINN and gated architecture. Since the architecture of the PECANN is not gated and uses a smooth hyperbolic tangent activation function, the model can only approximate roughly the C^0 solution at payoff. With this in mind, The RMSE for the Adaptive PECANN is plotted against $\hat{t}$ below in figure 11.

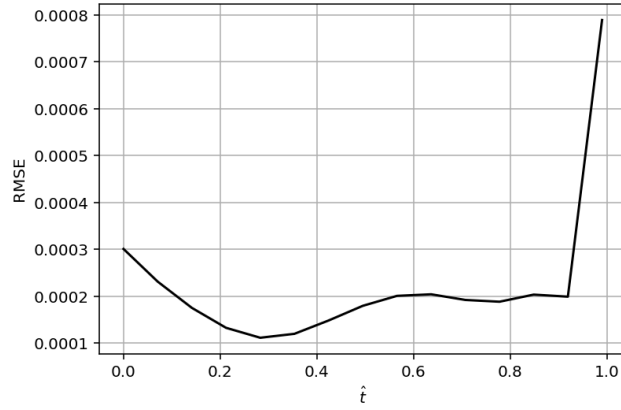


Figure 11: RMSE vs $\hat{t}$ for the Adaptive PECANN

The spike in RMSE as $\hat{t}$ approaches 1 (payoff) is clearly visible in figure 11 and expected given the architecture constraints. However, despite the peak RMSE of 0.0008 near payoff, the error of the adaptive PECANN is still approximately 280 times smaller than the PINN presented by Dhiman & Hu [7].

Despite this promising result from the PECANN, the current model has limitations. Firstly, only European style call options are considered here. The primary use case for a PECANN would be for more complex scenarios with many underlying assets, such as for rainbow options, or floating boundary conditions, such as for American options. In these cases, numerical solvers are necessary to approximate a solution since the closed-form method breaks down. Here a PECANN could offer superior computational efficiency over traditional numerical methods. Secondly, the model is only valid for a singular maturity of option, 1 year in this case, and must be retrained to predict the payoff function otherwise. Thirdly, assumptions such as constant volatility are made, and other important extensions of Black-Scholes, such as Heston volatility, have not yet been considered. Finally, gated architecture as discussed above has not been utilised therefore the solution at payoff is considerably less accurate than throughout the rest of the computational domain.

6.4 Seed-Variance Results

To ensure the network can repeatedly be trained and exhibit the same behaviour and accuracy of prediction as above, the same training process was repeated 10 times for each of the PINN, PECANN and adaptive PECANN models. The input data and collocation points were randomly drawn again for each run and the network initialised from different values. The training and validation processes were identical to those described and used previously in earlier sections. The resulting RMSE violin plots for each model type across the low, mid, and high range of volatilities and risk free rates are seen in the figures below.

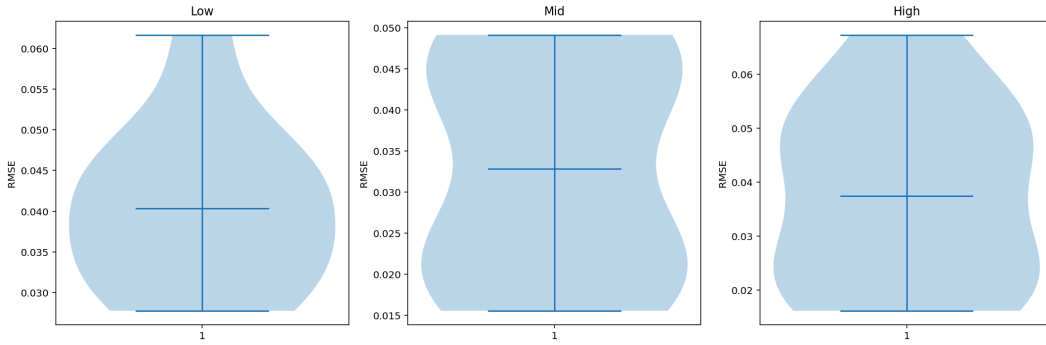


Figure 12: PINN RMSE Violin Plots

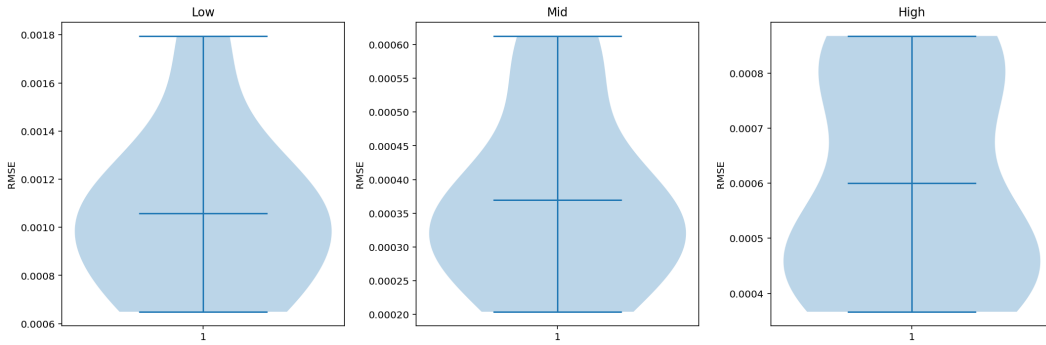


Figure 13: PECANN RMSE Violin Plots

The mean RMSE and a single standard deviation ($\overline{RMSE} \pm \sigma$) for each of the three models and parameter sets are provided in table 5.

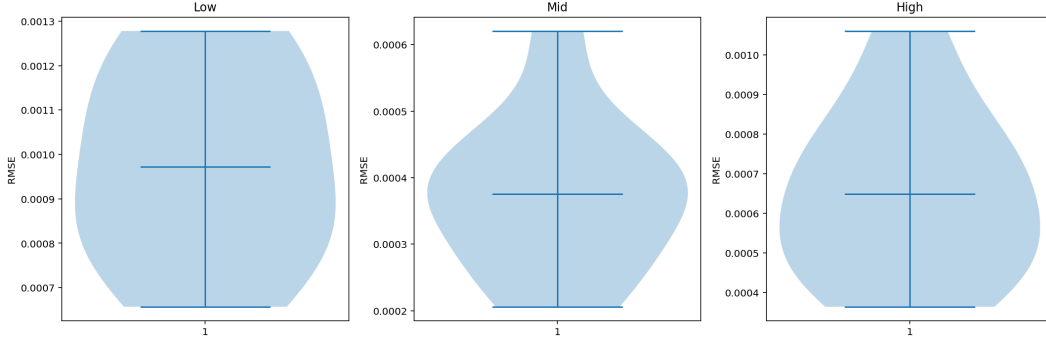


Figure 14: Adaptive PECANN RMSE Violin Plots

Model	Low RMSE	Mid RMSE	High RMSE	Training Time (s)
PINN	$(4.034 \pm 1.0) \times 10^{-2}$	$(3.284 \pm 1.4) \times 10^{-2}$	$(3.745 \pm 1.7) \times 10^{-2}$	175.4 $\pm$ 1.7
PECANN	$(1.057 \pm 0.3) \times 10^{-3}$	$(3.695 \pm 1.3) \times 10^{-4}$	$(6.001 \pm 2.0) \times 10^{-4}$	453.8 $\pm$ 6.7
Adaptive PECANN	$(9.719 \pm 2.2) \times 10^{-4}$	$(3.753 \pm 1.2) \times 10^{-4}$	$(6.485 \pm 2.1) \times 10^{-4}$	1331.2 $\pm$ 137.4

Table 5: Comparison of PINN, PECANN, and Adaptive PECANN Seed Variance Results at Present Day, $\hat{t} = 0$.

It should be noted that the larger training times listed in table 4 for the PINN and standard PECANN models is likely due to system load variations between the single isolated runs and multi-run seed-variance loops. The above plots and table 5 demonstrate that in addition to the previous promising adaptive PECANN result, when repeated over 10 different runs using different training data and starting collocation points, the average RMSE is only improved for the low set of parameters by a marginal amount. Upon inspection using Welch's t-test, The small differences in mean RMSE between the PECANN and adaptive PECANN are statistically indistinguishable results from noise with p-values of 0.48, 0.92 and 0.60 respectively for the low, mid and high parameters. The effect size of these results are small to negligible, with Cohen's d values of 0.32, 0.05, and 0.24 respectively, indicating any observed differences are small relative to the noise inherent in the training process. These results, paired with the factor of three increased training time for an adaptive over a standard PECANN, suggests that the standard PECANN is the favourable compromise of accuracy and computational cost in this study.

7 Debugging & Design Decisions

During the process of building the PINN and PECANN models, a variety of issues and challenges had to be overcome. For example, the choice of maximum penalty parameter, μ_{max} , seen in table 1 was vital alongside the careful selection of the growth rate, β , to prevent divergence of the penalty parameter. Initially a series of runs diverged with very large μ values creating a sizeable imbalance between the PDE, data, and boundary condition loss terms. For a similar reason a mean loss value was used in the overall loss function over a sum for the boundary condition and data terms. This again was in an effort to reduce the disparity in magnitudes of loss terms. Another issue faced was instability during the hand-off between the Adam and LBFGS optimisers. LBFGS using only Armijo conditions, without the strong Wolfe line search, is noted as unstable for complex loss landscapes such as PINNs or PECANNs [2]. This instability issue drove the selection of the strong Wolfe line search which allows for dynamic adjustment of the learning rate, promoting improved stability. The training epoch split of 4000 with Adam and 4000 with LBFGS was selected, compromising between stability, computational cost, and accuracy. LBFGS is sensitive to its starting point, and directly impacted by the number of epochs and convergence of the Adam optimiser in the initial training period [3]. Therefore, a variety of different splits were tested before making the final selection.

8 Future Work

As mentioned in the concluding remarks of the results section, the current PECANN model has a series of limitations and assumptions. Future improvements to this model should include: maturity generalisation, with the time to maturity, T , becoming a model input, gradient normalisation of the losses to improve the disparity between terms and so convergence in training, architecture changes to better capture the solution at payoff, an extension to American style or multi-asset options. Having made these improvements, the resulting model may be able to predict the payoff solution for a complex option with a single forward pass regardless of higher dimensionality, unlike any traditional method.

9 Conclusion

This paper has presented a Physics and Equality Constrained Artificial Neural Network (PECANN) for pricing European-style call options under a partially non-dimensional Black-Scholes equation, extended to accept risk-free rate and volatility as network inputs alongside asset price and time. Reformulating the standard PINN loss using the Augmented Lagrangian Method, so that boundary and terminal conditions are enforced as hard equality constraints rather than soft, equally-weighted loss terms, reduced boundary and data loss by approximately three orders of magnitude relative to a baseline PINN. The PECANN produced payoff curves visually indistinguishable from the closed-form solution across the tested range of volatilities and risk-free rates. Adaptive collocation refinement, concentrating additional training points in regions of high PDE residual, provided a small reduction in RMSE, particularly around $\hat{S} = 1$ where the payoff function has the highest curvature, and the resulting model achieves an RMSE approximately 280 times smaller than a comparable gated architecture from the Dhiman & Hu paper [7], but it must be noted that it is not an absolute comparison given the possible differences in training splits, costs and other parameters. The marginal improvement in accuracy of the adaptive PECANN was found to be statistically indistinguishable when compared to the standard PECANN. Therefore, the standard PECANN was found to be the optimal compromise of accuracy and computational cost.

These results support the case that the Augmented Lagrangian reformulation directly addresses the loss-imbalance failure mode identified in the standard PINN formulation, and that non-dimensionalisation combined with parametrising the network over r and σ is a viable route to a single model capable of pricing a continuous range of options rather than one fixed contract. The model's principal limitation is accuracy near the terminal payoff, where the smooth hyperbolic tangent activation struggles to represent the kink in the payoff function without a gated architecture.

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